

找一个函数. 使其仅在一点上连续

$$D(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases} \quad D(x) \text{ 处处不连续}$$

$$x \cdot D(x) = \begin{cases} x & x \in \mathbb{Q} \\ 0 & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

$$\lim_{x \rightarrow 0} x \cdot D(x) = 0 = D(0) \quad \Rightarrow \text{在 } x=0 \text{ 时连续}$$

$$\forall x_0 \neq 0, \lim_{x \rightarrow x_0} x \cdot D(x) \text{ 不存在. } \Rightarrow \text{在 } x \neq 0 \text{ 时间断}$$

$$R(x) = \begin{cases} 1, & x=0 \\ \frac{1}{p}, & x=\frac{q}{p} \in \mathbb{Q} \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases} \quad (\text{不可导})$$

$$\begin{cases} 1, & x=0 \\ \frac{1}{p^3}, & x=\frac{q}{p} \in \mathbb{Q} \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases} \quad (\text{部分可导})$$

$$\forall x_0, \lim_{x \rightarrow x_0} R(x) = 0 \quad \text{在无理点连续, 在有理点间断}$$

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } |R(x)| < \varepsilon, \forall 0 < |x - x_0| < \delta$$

$$\text{若 } R(x) \geq \varepsilon \Rightarrow x = \frac{q}{p} \in \mathbb{Q} \Rightarrow \frac{1}{p} \geq \varepsilon \Rightarrow \frac{1}{\varepsilon} \geq p$$

在 $(x_0 - 1, x_0 + 1)$ 中. 满足条件的有理数为有限个, 为 $x_1, x_2, \dots, x_k$

$$\text{取 } \delta = \min \{ |x_i - x_0| \} \text{ s.t. } |R(x)| < \varepsilon$$

(需要满足的 $\frac{1}{\varepsilon} \geq p$ 是有限的)

证明函数连续性

$$\text{1. } \lim_{x \rightarrow 0} a^x = 1 \quad (a > 0)$$

当 $x_0 = 0$ 时...

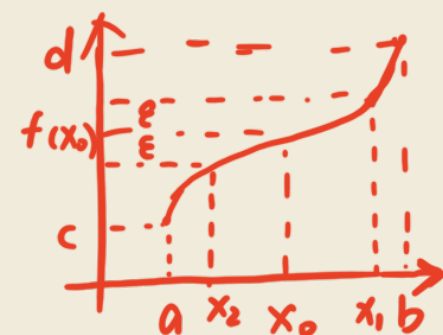
$$\text{当 } x_0 \neq 0 \text{ 时 } \lim_{x \rightarrow x_0} a^x = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta x = x - x_0}} a^{x_0 + \Delta x} = a^{x_0} \cdot \lim_{\Delta x \rightarrow 0} a^{\Delta x} = a^{x_0}$$

$$2 \quad \lim_{x \rightarrow 0} \sin x = 0 = \sin 0$$

$$\lim_{x \rightarrow x_0} \sin x = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta x = x - x_0}} \sin(x_0 + \Delta x) = \lim_{\Delta x \rightarrow 0} \sin x_0 \cdot \cos \Delta x + \cos x_0 \cdot \sin \Delta x = \sin x_0$$

反函数的连续性 $f: (a, b) \rightarrow (c, d)$ 是一一映射 $f(x)$ 严格单调

不妨设 f 严格单调 $\forall x_0 \in (a, b) \quad \forall \varepsilon > 0 \quad \varepsilon < \min \{d - f(x_0), f(x_0) - c\}$



$$\exists \delta \text{ s.t. } |f(x) - f(x_0)| < \varepsilon \quad \forall |x - x_0| < \delta$$

$$b > x_1 = f^{-1}(f(x_0) + \varepsilon) > x_0$$

$$a < x_2 = f^{-1}(f(x_0) - \varepsilon) < x_0$$

$$\text{取 } \delta = \min \{x_0 - x_2, x_1 - x_0\}$$

$$\forall |x - x_0| < \delta \text{ 时 } f(x) > f(x_0), f(x) < f(x_1) \quad |f(x) - f(x_0)| < \varepsilon$$

$$\lim_{x \rightarrow x_0} f(x) = f(x_0) \quad f(x) \text{ 在 } x = x_0 \text{ 上连续}$$

$$f^{-1}: (c, d) \rightarrow (a, b) \quad 1-1 \quad \text{严格单调}$$

由此 $y = a^x (a > 1) \quad \mathbb{R} \rightarrow (0, +\infty)$ 1-1 单调递增 则 $y = a^x$ 连续, 且反函数 $y = \log_a x$ 在 $(0, +\infty)$ 上单调

$$y = \sin x \quad [-\frac{\pi}{2}, \frac{\pi}{2}] \rightarrow [-1, 1]$$

$$y = \arcsin x \quad [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$$

$$\lim_{u \rightarrow b} f(u) = f(b) \quad \lim_{x \rightarrow x_0} g(x) = b$$

外函数连续

$$\text{则 } \lim_{x \rightarrow x_0} f(g(x)) = f(\lim_{x \rightarrow x_0} g(x)) = f(b)$$

$$\text{证: } \forall \varepsilon > 0 \quad |f(g(x)) - f(b)| < \varepsilon$$

$$\text{由 } \lim_{u \rightarrow b} f(u) = f(b) \text{ 知 } \exists \delta_1 > 0, \text{ s.t. } |f(u) - f(b)| < \varepsilon \quad \forall |u - b| < \delta_1$$

$$\text{对于 } \delta_1, \quad \lim_{x \rightarrow x_0} g(x) = b \quad \exists \delta > 0 \text{ s.t. } |g(x) - b| < \delta_1 \quad \forall |x - x_0| < \delta$$

计算 $\sqrt{x + \sqrt{x + \sqrt{x}}}$ 的阶 ($x \rightarrow 0$)

$$\text{设 } x^{\frac{1}{3}} = t.$$

$$\text{则有 } \lim_{t \rightarrow 0} \sqrt{t^8 + \sqrt{t^8 + \sqrt{t^8}}} = \lim_{t \rightarrow 0} \sqrt{t^8 + t^2 \sqrt{t^4 + 1}} = \lim_{t \rightarrow 0} t \sqrt{t^6 + 1} = t$$

$$\lim_{x \rightarrow 0} \frac{n \sqrt[n]{1+x^3} - 1}{x^3} = \lim_{x \rightarrow 0} \frac{(\sqrt[n]{1+x^3})^n - 1}{x^3 [(\sqrt[n]{1+x^3})^{n-1} + (\sqrt[n]{1+x^3})^{n-2} + \dots + 1]} = \frac{1}{n}$$

有理化

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{\log_a(1+x)}{x} = \log_a e$$

$$= \lim_{x \rightarrow 0} \log_a (1+x)^{\frac{1}{x}} = \log_a \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$$

$$\textcircled{2} \lim_{x \rightarrow 0} \frac{a^x - 1}{x} \quad \begin{array}{l} \text{令 } y = a^x - 1 \\ x = \log_a(1+y) \end{array} \quad \lim_{y \rightarrow 0} \frac{y}{\log_a(1+y)} = \ln a \quad (\text{描述同一极限过程})$$

利用ln化乘除

$$\textcircled{3} \lim_{x \rightarrow 0} \frac{(1+x)^u - 1}{x} = \frac{e^{u \ln(1+x)} - 1}{x} = \frac{u \ln(1+x)}{x} = u$$

$$\textcircled{4} \lim_{x \rightarrow 0} \frac{\arcsin x}{x} = \lim_{y \rightarrow 0} \frac{y}{\sin y} = 1$$

$$\lim_{x \rightarrow 0} \frac{\arctan x}{x} = \lim_{y \rightarrow 0} \frac{y}{\tan y} = 1$$

$$\textcircled{5} \lim_{x \rightarrow \infty} \left(1 + \frac{\alpha}{x}\right)^x = \lim_{x \rightarrow \infty} e^{\frac{\ln(1 + \frac{\alpha}{x})}{\frac{1}{x}}} = \lim_{x \rightarrow 0} e^{\frac{\ln(1 + \alpha x)}{x}} = e^\alpha$$

$$\text{若为 } \left(1 + \frac{\alpha}{x} + \frac{\beta}{x^2}\right)^x? \quad \lim_{x \rightarrow \infty} \frac{\ln(1 + \alpha x + \beta x^2)}{x} = e^\alpha.$$

$$\textcircled{6} \lim_{x \rightarrow x_0} u(x)^{v(x)} = e^{\lim_{x \rightarrow x_0} v(x) \cdot \ln \lim_{x \rightarrow x_0} u(x)} = e^{v_0 \cdot \ln u_0} = u(x_0)^{v(x_0)} \quad (\text{要求 } u(x), v(x) \text{ 连续})$$

$$\textcircled{7} \lim_{x \rightarrow 0} (\cos x)^{\cot^2 x} = \lim_{x \rightarrow 0} e^{\frac{\ln \cos x}{\tan^2 x}} = \lim_{x \rightarrow 0} e^{\frac{\ln[1 - (1 - \cos x)]}{\tan^2 x}} = e^{\frac{-1 - \cos x}{x^2}} = e^{-\frac{1}{2}}$$

$$\textcircled{8} \lim_{x \rightarrow 0} (e^x - x)^{\frac{1}{\sin x}} = \lim_{x \rightarrow 0} e^{\frac{\ln(e^x - x)}{\sin x}} = \lim_{x \rightarrow 0} e^{\frac{e^x - x - 1}{x}} = 1$$

$$\lim_{x \rightarrow 0} \frac{\log_a(1+x)}{x} = \log_a e \quad (x \rightarrow 0)$$

$$\ln(1+x) \sim x \quad (x \rightarrow 0)$$

$$e^x - 1 \sim x \quad (x \rightarrow 0) \quad \leftarrow$$

$$(1+x)^u - 1 \sim ux \quad (x \rightarrow 0)$$

$$\arcsin x \sim x \quad (x \rightarrow 0)$$

$$\arctan x \sim x \quad (x \rightarrow 0)$$

$$a^x - 1 \sim x \ln a \quad (a^x = e^{x \ln a})$$

① $\lim_{x \rightarrow \frac{\pi}{4}} (\tan x)^{\frac{1}{\cos x - \sin x}} \stackrel{\Delta}{=} y = x - \frac{\pi}{4} \quad \lim_{y \rightarrow 0} e^{\frac{\ln \frac{1+\tan y}{1-\tan y}}{-\sqrt{2} \cdot \sin y}} = e^{\frac{\frac{2y}{1-\tan y}}{-\sqrt{2}y}} = e^{-\sqrt{2}}$ 换元使其趋于0.

设 $a, b \in \mathbb{R}^+$ 证明 $x = a \sin x + b$ 至少有一个不超过 $a+b$ 的正根

$f(x) = x - (a \sin x + b)$ 在 $[0, a+b]$ 上连续

若 $f(a+b) = 0$ 则 $x = a+b$ 是所求目标 x .

若 $f(a+b) > 0$ 则有 $f(0) \cdot f(a+b) < 0$. 由零点定理知 $\exists \varepsilon \in (0, a+b)$ s.t. $f(\varepsilon) = 0$, 则 ε 即为目标. 所求.

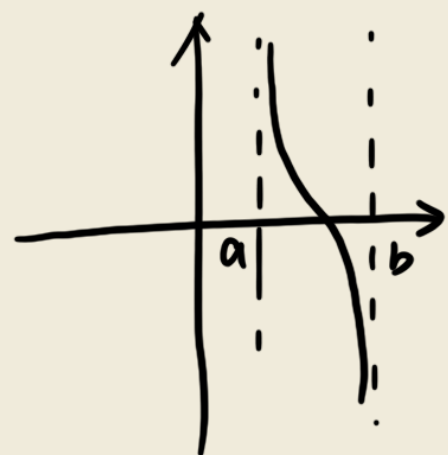
方程 $f(x) = \frac{\sin x}{x} - \frac{99}{100}$

函数为基本函数, 函数连续.

$$\left. \begin{array}{l} \lim_{x \rightarrow 0^+} f(x) = \frac{1}{100} > 0 \\ \lim_{x \rightarrow \pi} f(x) = -\frac{99}{100} < 0 \end{array} \right\} \text{存在 } x_0 \in (0, \pi] \text{ s.t. } \lim_{x \rightarrow x_0} f(x) = \frac{1}{200} > 0$$

$$\Rightarrow \exists \varepsilon \in [x_0, \pi] \text{ s.t. } f(\varepsilon) = 0$$

若函数如图



函数连续

$$\lim_{x \rightarrow a^+} = +\infty$$

$$\text{则 } \exists x_1 \text{ s.t. } f(x_1) = 1$$

$$\lim_{x \rightarrow b^-} = -\infty$$

$$\exists x_2 \text{ s.t. } f(x_2) = -1$$

$$\Rightarrow \exists \varepsilon \in [x_1, x_2] \text{ s.t. } f(\varepsilon) = 0$$

函数 $f(x)$ 在开区间 (a, b) 内连续, 且 $x_1, x_2, \dots, x_n \in (a, b)$

又 $t_i > 0$ ($i = 1, 2, \dots, n$) 且 $\sum_{i=1}^n t_i = 1$

试证 $\exists \xi \in (a, b)$, s.t. $f(\xi) = \sum_{i=1}^n t_i f(x_i)$

记 $M = f(y_1) = \max_{x \in [x_1, x_n]} f(x)$, $m = f(y_2) = \min_{x \in [x_1, x_n]} f(x)$

$$g(y_1) = f(y_1) - \sum_{i=1}^n t_i f(x_i) = \sum_{i=1}^n t_i (f(y_1) - f(x_i)) \geq 0$$

$$g(y_2) = f(y_2) - \sum_{i=1}^n t_i f(x_i) = \sum_{i=1}^n t_i (f(y_2) - f(x_i)) \leq 0$$

若 $g(y_1) = 0$ 取 $\xi = y_1$, $g(y_2) = 0$, 取 $\xi = y_2$

否则 $g(y_1)g(y_2) < 0$ 由零点定理, $\exists \xi \in [y_1, y_2] \text{ 或 } [y_2, y_1] \subset [x_1, x_2]$ s.t. $g(\xi) = 0$

